The Space Station Swindle

Whodunnit? · Systems of Equations · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Systems of Equations	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Solving systems of linear equations by substitution, elimination and from word problems.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Engineer Mara took it, in the Airlock, using this method: Launched in a supply capsule.

The reveal (read aloud): While the countdown ticked, **Engineer Mara** slipped into the **Airlock**, tucked the Star Crystal into a **supply capsule**, and launched it toward her own hidden spacecraft. Every alibi and method you cleared led to her. Out-of-this-world detective work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Commander Vega · Place: Command Deck
2	Method: Hidden in a cargo pod
3	Suspect: Dr. Okoye · Place: Greenhouse Dome
4	Method: Wrapped in a spacesuit
5	Suspect: Pilot Zhang · Place: Star Lounge
6	Method: Packed in a food tray
7	Suspect: Cadet Sol · Place: Engine Room
8	Method: Disguised as a moon rock
9	Suspect: Chef Bao · Place: Galley
10	Method: Floated in a zero-g duffel

Answer Key

The Space Station Swindle · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Solve the system: $y = 2x - 1$ and $y = x + 1$	**A.** (2, 3)	$2x - 1 = x + 1$ $x = 2$, then $y = 3$ (2, 3)	Commander Vega was on the Command Deck running the countdown. The flight recorder confirms it.
2	Solve the system: $y = -x + 5$ and $y = 2x - 4$	**B.** (3, 2)	$-x + 5 = 2x - 4$ $9 = 3x$, so $x = 3$ and $y = 2$ (3, 2)	Every cargo pod was scanned. All of them are filled with spare parts.
3	Solve the system: $y = -x + 1$ and $y = x + 5$	**C.** (−2, 3)	$-x + 1 = x + 5$ $-4 = 2x$, so $x = -2$ and $y = 3$ (−2, 3)	Dr. Okoye was watering moon-lettuce in the Greenhouse Dome. The sensor logged her entry.
4	Solve the system: $2x + y = 4$ and $x - y = 5$	**D.** (3, −2)	add the equations: $3x = 9$ $x = 3$, then $y = -2$ (3, −2)	The spacesuits were all in their lockers, sealed. Nothing was hidden in them.
5	Solve the system: $3x + 2y = -5$ and $x + y = -1$	**C.** (−3, 2)	$y = -1 - x$; $3x + 2(-1 - x) = -5$ $x - 2 = -5$, so $x = -3$ and $y = 2$ (−3, 2)	Pilot Zhang was gazing at Earth in the Star Lounge. The cameras caught him there.
6	Solve the system: $x + 2y = -4$ and $3x - y = 9$	**D.** (2, −3)	$y = 3x - 9$; $x + 2(3x - 9) = -4$ $7x = 14$, so $x = 2$ and $y = -3$ (2, −3)	The food trays were counted. None had been tampered with.
7	Solve the system: $2x - 3y = 5$ and $4x + y = -11$	**A.** (−2, −3)	multiply the second by 3 and add: $14x = -28$ $x = -2$, then $y = -3$ (−2, −3)	Cadet Sol was scrubbing valves in the Engine Room, supervised by two crew members.
8	Two numbers x and y add to -5. The first minus twice the second equals 1. Find the ordered pair (x, y).	**B.** (−3, −2)	$x + y = -5$ and $x - 2y = 1$ subtract: $3y = -6$, so $y = -2$ and $x = -3$ (−3, −2)	The “moon rock” on the shelf is just painted foam, so no disguise was used.
9	4 pens and 1 notebook cost \$9. 1 pen and 4 notebooks cost \$6. Let x = price of a pen and y = price of a notebook. Find (x, y).	**D.** (2, 1)	$4x + y = 9$ and $x + 4y = 6$ $16x + 4y = 36$; subtract: $15x = 30$ $x = 2$ and $y = 1$ (2, 1)	Chef Bao was heating dinner trays in the Galley. The oven log proves it.
10	Solve the system: $y = 3x - 1$ and $2x + y = 4$	**C.** (1, 2)	$2x + (3x - 1) = 4$ $5x = 5$, so $x = 1$ and $y = 2$ (1, 2)	The zero-g duffel bags were checked. None held the Crystal.

Wrong-Answer Map

The Space Station Swindle · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	B. (3, 2)	swapped x and y	Engineer Mara was repotting seedlings in the Greenhouse Dome. A soil-moisture sensor logged her all afternoon.
1	C. (0, 1)	sign slip: 2x−1=x+1 gives x=0	Cadet Sol was checking suit seals at the Airlock. The door counter clicked when he walked in.
1	D. (2, 1)	used y=x−1	The supply capsule bay was inspected. Its hatch was welded shut and never opened.
2	A. (1, 4)	arithmetic slip: 3x=3	Chef Bao was hauling a trash bag to the Airlock for disposal. The cargo log shows his visit.
2	C. (3, −2)	sign error in y	No supply capsule left the station today. The launch rail shows zero activity.
2	D. (2, 3)	swapped x and y	Engineer Mara was stargazing in the Star Lounge with the whole night shift. Everyone remembers her telescope jokes.
3	A. (3, −2)	swapped x and y	Engineer Mara was tightening bolts in the Engine Room. Her wrench is still in the pump housing, tagged with the time.
3	B. (2, −1)	sign error on x	Dr. Okoye was collecting dust samples at the Airlock. Her sample tags are stamped with the time.
3	D. (−2, −3)	sign error on y	The capsule launch tubes are covered in dust. Nobody has loaded a capsule in weeks.
4	A. (3, 2)	sign error on y	Pilot Zhang was suiting up for a spacewalk at the Airlock. The suit log lists his name.
4	B. (1, 2)	arithmetic slip: 3x=3	A scan of the supply capsules found them all sealed and still sitting on their racks.
4	C. (−2, 3)	swapped x and y	Engineer Mara was helping with the countdown on the Command Deck. Her badge swipe shows she never left.
5	A. (2, −3)	swapped x and y	Engineer Mara was stirring a pot of stew in the Galley. The cook says she asked for seconds.
5	B. (3, −4)	sign error on x	Commander Vega was inspecting the outer door at the Airlock. Two crew members walked the tour with her.
5	D. (−3, −2)	sign error in y=−1−x	The supply capsule's cargo scale reads zero. Nothing heavier than a sandwich went in.
6	A. (−2, −15)	sign error on x	The supply capsule's launch log is blank today. No countdown was ever started for it.
6	B. (2, 3)	sign error on y	Chef Bao was mopping the floor of the Airlock. The wet-floor sign has his name on it.
6	C. (−3, 2)	swapped x and y	Engineer Mara was playing zero-g chess in the Star Lounge. Her opponent still has the board to prove it.
7	B. (−2, 3)	sign error on y	The supply capsule was weighed before and after the shift. The numbers match exactly.
7	C. (2, −19)	sign error on x	Dr. Okoye was calibrating a sensor at the Airlock. The sensor tag has the time written on it.
7	D. (−3, −2)	swapped x and y	Engineer Mara was trimming vines in the Greenhouse Dome. A time-lapse camera caught her there.
8	A. (−3, 2)	sign error on y	The supply capsule's door is jammed with frost. It could not have been opened today.

8	C. (−2, −3)	swapped x and y	Engineer Mara was unpacking snack boxes in the Galley. The inventory sheet has her initials.
8	D. (−7, 2)	sign error on y	Commander Vega was reading pressure dials at the Airlock during the safety check. The dial log matches.
9	A. (1, 2)	swapped x and y	Engineer Mara was oiling a pump in the Engine Room. The pump's maintenance chip has her name on it.
9	B. (1, 5)	arithmetic slip in x	The supply capsule's tracking beacon never moved from its dock. It stayed put all day.
9	C. (2, −1)	sign error on y	Cadet Sol was sweeping bolts off the floor of the Airlock. His cleaning tablet recorded every minute.
10	A. (2, 1)	swapped x and y	Engineer Mara was calling out numbers on the Command Deck during the final systems check. The recorder has her voice.
10	B. (1, 4)	used y=3x+1	Pilot Zhang was tightening a hatch wheel at the Airlock. His tool belt scanner confirmed it.
10	D. (5, 14)	forgot to divide by 5	Mission control checked the supply capsule's manifest. It lists only water and socks.